BIOTA1

Mobile Application

Product Requirements Document

Document Version	1.0 — Initial Release
Date	March 2026
Platform	Android & iOS (Flutter WebView)
Web App URL	https://biota1.lovable.app/
Status	Draft — Ready for Development

1. Executive Summary

Biota1 is a web application currently hosted at https://biota1.lovable.app/ and built using Lovable, a React-based rapid development platform. This document defines the requirements for wrapping that existing web application into native Android and iOS mobile apps using Flutter's WebView package.

The approach is a WebView wrapper — the app loads the live web URL inside a native shell. This gives users a native app experience (splash screen, app icon, home screen presence, store distribution) without rebuilding the frontend.

The primary deliverable from a development standpoint is a Flutter project that handles: splash screen display, WebView loading, connectivity handling, and native OS integration — while all business logic and UI continue to live on the web.

2. Project Scope

2.1 In Scope

- Flutter project setup targeting Android and iOS
- Branded splash screen with Biota1 logo
- Full-screen WebView loading https://biota1.lovable.app/
- Internet connectivity detection and error screen
- Android APK and App Bundle (.aab) build
- iOS IPA build (requires macOS + Apple Developer account)
- App icon configuration for both platforms
- Basic WebView settings: JavaScript enabled, local storage, cookies

2.2 Out of Scope

- Rebuilding any UI/UX in Flutter natively
- Backend or API changes
- Push notifications (can be added in v2)
- Offline mode / content caching
- In-app purchases
- Analytics SDK integration (can be added in v2)

3. Technical Stack

Component	Specification
Framework	Flutter (latest stable — 3.x)
Language	Dart
WebView Package	webview_flutter ^4.x (official Flutter package)
Connectivity	connectivity_plus ^5.x
Splash Screen	flutter_native_splash ^2.x
App Icon	flutter_launcher_icons ^0.13.x
Android Min SDK	API 21 (Android 5.0 Lollipop)
Android Target SDK	API 34 (Android 14)
iOS Min Version	iOS 13.0
Web Target URL	https://biota1.lovable.app/

4. Splash Screen

4.1 Purpose

The splash screen is displayed immediately when the app is launched, while the WebView initialises and the web page begins loading. It serves as the brand entry point and masks any loading delay.

4.2 Design Requirements

Property	Requirement	Notes
Background colour	White (#FFFFFF) or brand colour — confirm with designer	Must match logo background
Logo	Biota1 logo centred vertically and horizontally	Provided by client
Logo size	40% of screen width, max 240dp	Scale proportionally
Logo format	PNG with transparent background, minimum 1024×1024px	Higher res for crisp rendering
Duration	Displayed until WebView finishes loading, minimum 1.5 seconds	Do not dismiss early
Animation	Optional: subtle fade-in on logo (200ms ease-in)	Keep it professional
Status bar	Hidden during splash, restored after	Immersive entry
Orientation	Portrait only during splash	Consistent entry point

4.3 Implementation — flutter_native_splash

Use the flutter_native_splash package to generate true native splash screens for both platforms. This ensures the splash appears before Flutter engine initialises (no white flash).

pubspec.yaml configuration:

```
flutter_native_splash:
  color: "#FFFFFF"
  image: assets/images/logo.png
  android_12:
    image: assets/images/logo.png
    color: "#FFFFFF"
  ios: true
  fullscreen: true
```

Run after configuration:

```
dart run flutter_native_splash:create
```

4.4 Splash-to-WebView Transition

1. App launches — native splash screen shows immediately (OS-level).
2. Flutter engine initialises — programmatic splash (if any) shown.
3. WebView widget is created and begins loading the URL in background.
4. WebView fires onPageFinished callback.
5. Wait for minimum 1.5 seconds from app start (whichever is later).
6. Fade splash out (opacity animation, 300ms).
7. Reveal WebView at full opacity.

5. App Icon

5.1 Requirements

Platform	Requirement	Format
Android	1024×1024px adaptive icon (foreground + background layers)	PNG
Android 12+	Adaptive icon with safe zone: logo within inner 66% of canvas	PNG
iOS	1024×1024px, no transparency, no rounded corners (Apple applies mask)	PNG
Play Store	512×512px feature graphic	PNG or JPEG
App Store	1024×1024px	PNG

5.2 flutter_launcher_icons Configuration

```
flutter_launcher_icons:
  android: "launcher_icon"
  ios: true
  image_path: "assets/images/icon.png"
  adaptive_icon_background: "#FFFFFF"
  adaptive_icon_foreground: "assets/images/icon_foreground.png"
  min_sdk_android: 21
```

6. WebView Requirements

6.1 Core Settings

Setting	Value / Behaviour
Initial URL	https://biota1.lovable.app/
JavaScript	Enabled
DOM Storage	Enabled (localStorage, sessionStorage)
Cookies	Enabled and persisted across sessions
Mixed content	Allowed (HTTPS page may load HTTP resources if needed)
Zoom controls	Disabled (site is responsive, no zoom needed)
Scrolling	Native scrolling enabled
Background colour	White (#FFFFFF) — shown while page loads
User agent	Default Flutter WebView UA (do not override)
Cache mode	LOAD_DEFAULT — use HTTP cache headers from server
Geolocation	Request permission only if the website requests it
File access	Disabled
Content URL access	Disabled

6.2 Navigation Behaviour

- All navigation within https://biota1.lovable.app/* — load inside WebView.
- External links (different domain) — open in system browser (url_launcher).
- mailto: links — open system email client.
- tel: links — open system dialler.
- Android hardware back button — navigate WebView back history. If no history remains, show exit confirmation dialog.

6.3 WebView Controller Setup (Dart pseudocode)

```
late final WebViewController _controller;

_controller = WebViewController()
  ..setJavaScriptMode(JavaScriptMode.unrestricted)
  ..setBackgroundColor(Colors.white)
  ..setNavigationDelegate(NavigationDelegate(
    onPageStarted: (_) => setState(() => _loading = true),
    onPageFinished: (_) => _onLoaded(),
    onWebResourceError: (_) => _showError(),
    onNavigationRequest: (req) => _handleNav(req),
  ))
  ..loadRequest(Uri.parse('https://biota1.lovable.app/'));
```

7. Connectivity & Error States

7.1 No Internet Screen

When the device has no internet connection at launch or during use, display a branded error screen instead of a blank WebView.

Element	Content	Notes
Logo	Biota1 logo, greyed out slightly	opacity: 0.6
Icon	WiFi-off icon (Material Icons)	Centred, 64dp, brand colour
Title	"No Internet Connection"	Bold, 20sp
Body text	"Please check your connection and try again."	14sp, grey
CTA button	"Retry"	Brand colour, full width
Retry action	Re-check connectivity, reload WebView if online	

7.2 Page Load Error Screen

If the WebView fails to load the page (server error, timeout, DNS failure), display:

- Biota1 logo
- Error message: "Could not load the page. Please try again."
- Retry button that calls controller.reload()

7.3 Connectivity Monitoring

- Use connectivity_plus to listen for network changes.
- When connection is restored from offline state, automatically trigger a reload.
- Show a brief snackbar: "Back online" when connection restores.

8. Android-Specific Requirements

8.1 Manifest Permissions

```
<uses-permission android:name="android.permission.INTERNET"/>
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>
<uses-permission android:name="android.permission.ACCESS_WIFI_STATE"/>
```

8.2 Android Configuration

Setting	Value
Application ID	com.biota1.app (confirm with team)
App label	Biota1
Min SDK	21
Target SDK	34
Compile SDK	34
Version name	1.0.0
Version code	1
android:usesCleartextTraffic	false (HTTPS only)
Orientation	portrait (set in AndroidManifest)
Hardware acceleration	true (required for WebView)
Build output	App Bundle (.aab) for Play Store, APK for testing

8.3 Back Button Handling

```
WillPopScope(
  onWillPop: () async {
    if (await _controller.canGoBack()) {
      _controller.goBack();
      return false;
    }
    // Show exit dialog
    return await _showExitDialog();
  },
)
```

9. iOS-Specific Requirements

Setting	Value
Bundle Identifier	com.biota1.app (confirm with team)
Display Name	Biota1
Minimum iOS	13.0
Orientation	Portrait only
NSAppTransportSecurity	Not required (HTTPS only)
WKWebView	Used automatically by webview_flutter on iOS
Swipe to go back	Enabled — mirrors iOS Safari behaviour

Note: iOS builds require a Mac with Xcode installed and a paid Apple Developer Program membership ($99/year) for App Store distribution.

10. Flutter Project Structure

```
biota1_app/
├── android/                    # Android native project
├── ios/                        # iOS native project
├── assets/
│   └── images/
│       ├── logo.png            # Splash screen logo
│       ├── icon.png            # App icon (1024×1024)
│       └── icon_foreground.png  # Adaptive icon foreground
├── lib/
│   ├── main.dart               # Entry point
│   ├── app.dart                # MaterialApp setup
│   ├── screens/
│   │   ├── splash_screen.dart
│   │   ├── webview_screen.dart
│   │   └── no_internet_screen.dart
│   └── utils/
│       └── connectivity_helper.dart
├── pubspec.yaml                # Dependencies & assets
└── flutter_native_splash.yaml # Splash config
```

11. Full pubspec.yaml

```
name: biota1_app
description: Biota1 Mobile Application
version: 1.0.0+1

environment:
  sdk: '>=3.0.0 <4.0.0'

dependencies:
  flutter:
```

```
    sdk: flutter
  webview_flutter: ^4.4.2
  connectivity_plus: ^5.0.2
  url_launcher: ^6.2.4

dev_dependencies:
  flutter_test:
    sdk: flutter
  flutter_native_splash: ^2.3.9
  flutter_launcher_icons: ^0.13.1

flutter:
  uses-material-design: true
  assets:
    - assets/images/
```

12. Screen Flow & User Journey

Step	Screen / Action	Trigger
1	Native splash (OS-level) shows immediately	App launch
2	Flutter initialises, programmatic splash shown, WebView loads in background	Engine ready
3a	Splash dismissed → WebView revealed	Page loaded AND 1.5s elapsed
3b	No internet screen shown	No connectivity detected at step 2
3c	Error screen shown	WebView onWebResourceError fires
4	User interacts with Biota1 web app inside WebView	Normal use
5	User presses Android back → WebView navigates back	Hardware back button
6	No back history → Exit confirmation dialog	Back button at root
7	Connection lost mid-session → snackbar "No internet"	connectivity_plus stream
8	Connection restored → auto-reload WebView	connectivity_plus stream

13. Assets Required from Client

The following files must be provided before development can begin:

Asset	Specification	Status
logo.png (splash)	PNG, transparent bg, min 1024×1024px	Provided by client
icon.png (app icon)	PNG, no transparency, 1024×1024px	Provided by client
icon_foreground.png	Logo on transparent bg, logo within inner 66% of 1024×1024px canvas	Provided by client
Splash bg colour (hex)	Hex colour code to match brand	Confirm with client
App name (display)	Name shown under icon on home screen	"Biota1"

14. Acceptance Criteria

14.1 Splash Screen

- Splash screen appears immediately on cold start with zero white flash.
- Biota1 logo is visible, centred, and correctly sized.
- Splash is not dismissed before 1.5 seconds.
- Splash is dismissed only after the web page has loaded.
- Fade transition to WebView is smooth (no abrupt jump).

14.2 WebView

- https://biota1.lovable.app/ loads correctly and is fully functional.
- JavaScript works (site relies on React — JS must be enabled).
- No horizontal overflow or zoom issues.
- Internal links navigate within the WebView.
- External links open in the system browser.

14.3 Connectivity

- No internet at launch shows the branded error screen.
- Retry button successfully reloads when connection is restored.
- Loss of connection mid-session shows snackbar notification.
- App auto-reloads when connection is restored after being lost.

14.4 Android

- Back button navigates WebView history.
- Back button at root shows exit dialog.
- App installs and runs on Android 5.0+ (API 21+).

- Portrait orientation is locked.
- App icon displays correctly on launcher.

15. Build & Release Commands

15.1 Setup

```
flutter pub get
dart run flutter_native_splash:create
dart run flutter_launcher_icons
```

15.2 Android

```
# Debug APK (for testing)
flutter build apk --debug

# Release APK (for direct install)
flutter build apk --release

# App Bundle for Play Store
flutter build appbundle --release
```

15.3 iOS (requires macOS + Xcode)

```
flutter build ios --release
# Then archive via Xcode → Product → Archive
```

16. Future Enhancements (v2)

Feature	Description	Priority
Push Notifications	Firebase Cloud Messaging integration for user alerts	High
Deep Linking	Open specific pages in-app via URL scheme	High
Analytics	Firebase Analytics or Mixpanel for user behaviour tracking	Medium
Biometric Auth	Face ID / fingerprint lock on app open	Medium
Pull to Refresh	Native gesture to reload WebView	Low
Share Sheet	Share current page URL via native share dialog	Low
Landscape support	Allow landscape on tablet devices	Low

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